

R E S E R V

A purpose-built booking system for the Siebel Center for Design.

Built in-house to replace Waitwhile — students join from Discord, staff manage from a web dashboard.

Ash Castelino · Project Lead

Siebel Center for Design · Agentic AI @ UIUC · May 2026

Waitwhile costs money — and wasn't built for SCD.

What waitwhile lacks

Waitwhile is a **generic SaaS queue tool**. It charges per seat, doesn't know about Discord, and gives staff almost nothing about how the shop is actually being used.

The stake

Idle machines and ghost reservations cost the shop real time — every week.



Students forget to check the queue.

No-shows tie up machines for hours, and the next person in line has no way to know.



Staff have no visibility.

No view of who's using what, who failed verification, or what the trends are.



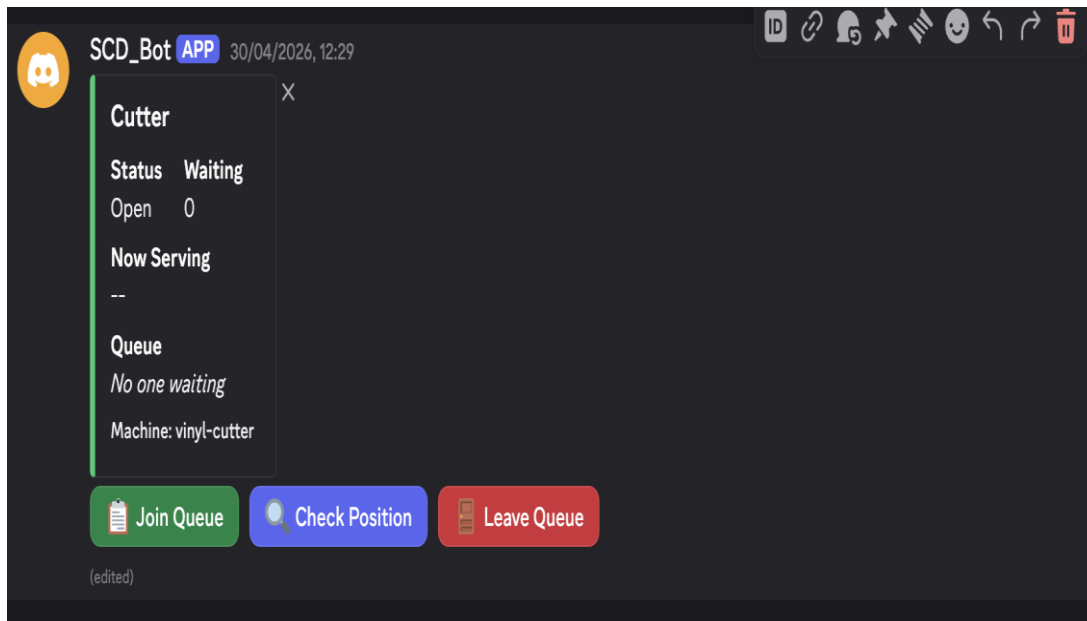
No identity check.

Anyone could fill the queue — including students from other campuses or non-students entirely.

Two surfaces. One system.

STUDENTS LIVE HERE

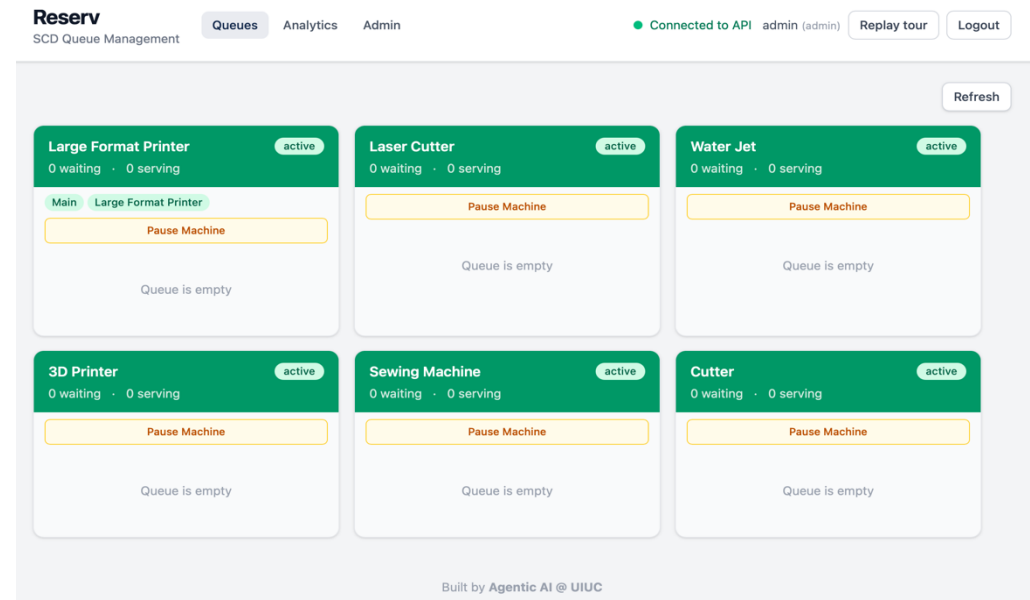
Discord



Three buttons. The channel they already live in.

STAFF LIVE HERE

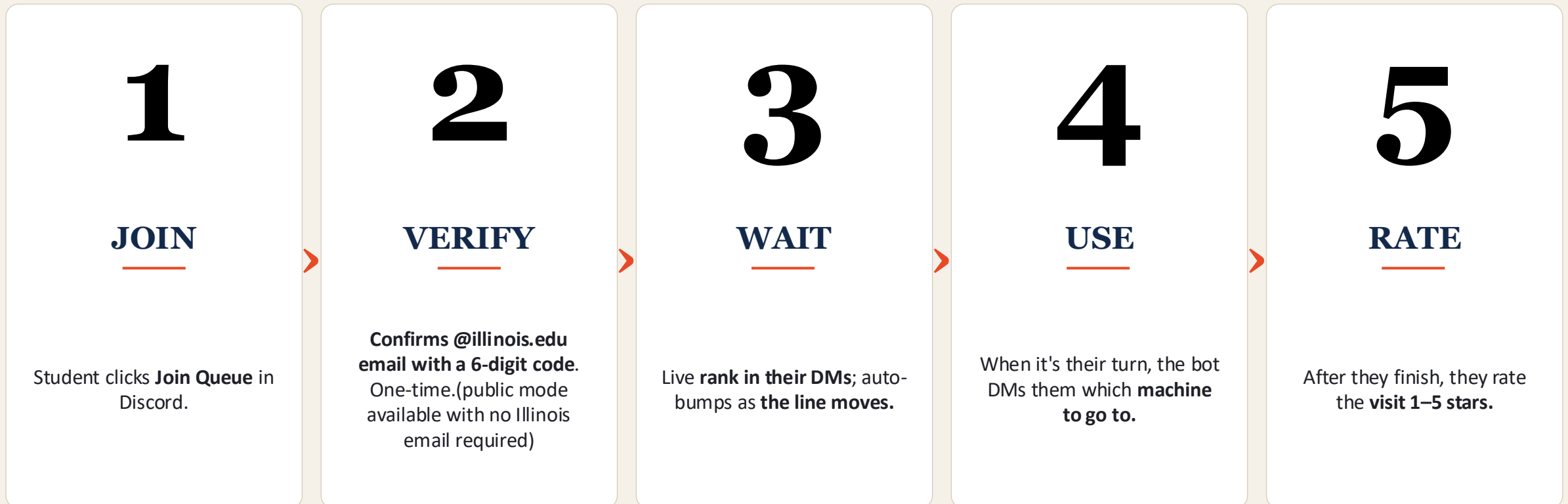
Web Dashboard



Live machine columns. One screen. One source of truth.

One join button in Discord. One dashboard for staff. Everything in between is automatic.

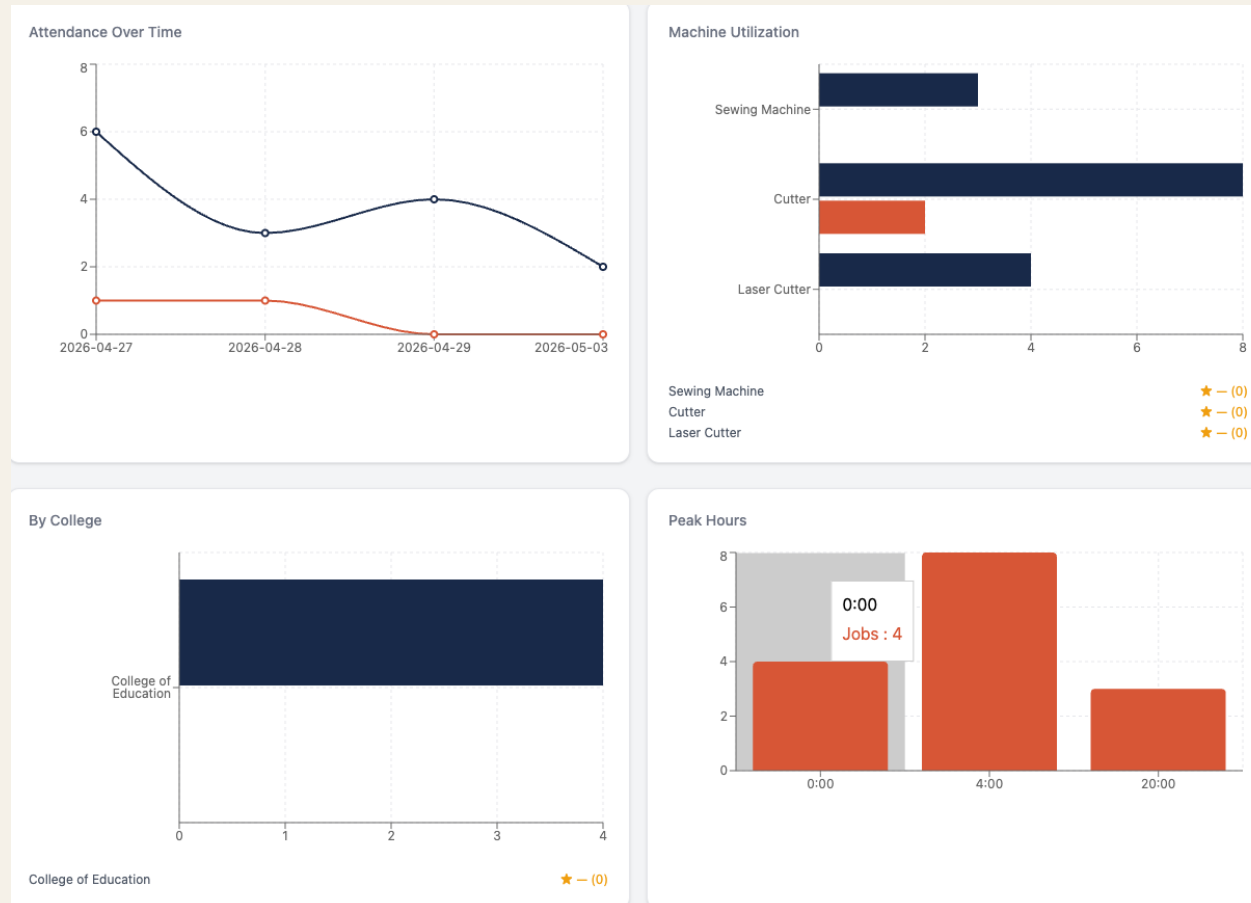
The student journey, end to end.



No app to download. No new account. They already have Discord.

An analytics dashboard built for the admin.

ANALYTICS DASHBOARD



1

Live queue across **all machines on one screen.**

2

One-click overrides — bump, skip, mark complete.

3

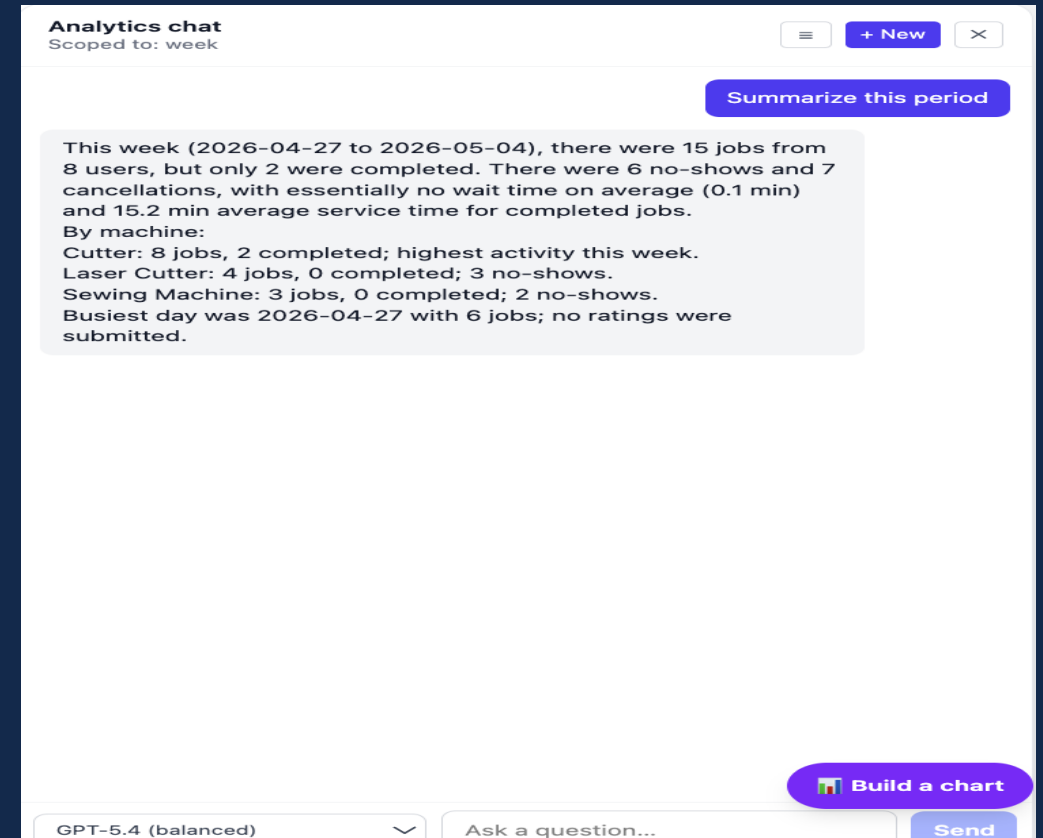
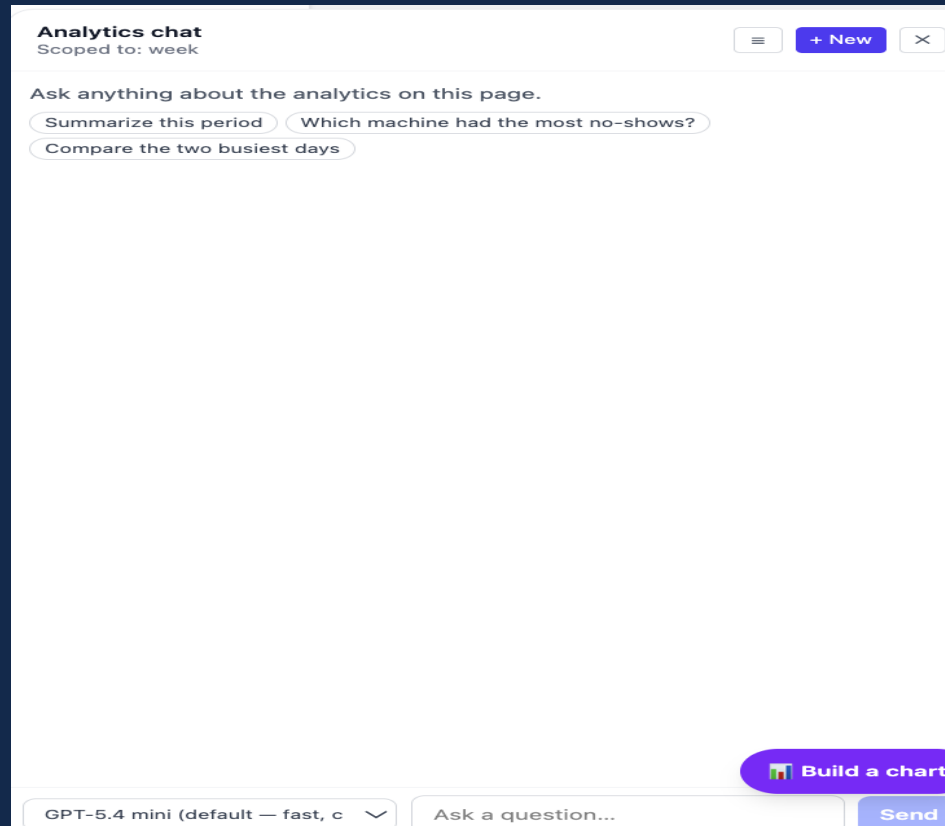
Per-college and per-machine filters — engineering students on the laser cutter this week?

4

AI-written daily summary — plain English, no spreadsheet diving.

Talk to your data.

STAFF



WHY IT MATTERS

Staff ask in plain English. The system pulls the answer, builds a chart, lets them pin it. Decisions in seconds — not week-end reports.

We thought of everything. Configurable, no code changes.

1 Multi-unit machines(editable)

One Laser Cutter row, three physical machines, three users served at once.

2 Maintenance mode

Per-machine. Banner shows publicly so students know before they queue.

3 Public-event mode

Drops email verification for one-off open houses and demo days.

4 Reminder + grace

Admin sets a 30-min reminder, 10-min grace before auto-finishing the slot.

The lab changes; the system follows — without us shipping new code.

What it costs vs. what it replaces.

WAITWILE

\$X / mo

- Generic SaaS — not Discord-aware.
- Per-seat fees grow with usage.
- No way to query the data in plain English.
- Branded SaaS plans for any custom features.

RESERV

\$6 / mo

- Runs on a \$6/mo VPS — flat cost.
- Tailored to SCD; Discord-native flow.
- AI analytics built in, not an upsell.
- No per-seat, no third-party integration fees.

Honest trade-off: we maintain it — but it's a single Python process and one SQLite file.

Production-ready, running today.

6+

machines tracked

Across the SCD floor.

305

automated tests passing

CI green on every push.

25+

API endpoints

Queue, analytics, chat, admin.

120

students onboarded

And growing weekly.

4.8

avg post-visit rating (★)

From the in-Discord ratings.

Production-ready. Running in the shop right now.

Where we go, and what we need from you.

ROADMAP

Next semester

- 1 Per-physical-unit queues**
Let students pick which laser cutter they want.
- 2 Calendar + booking windows**
For high-demand machines that need scheduled slots.
- 3 Slack mirror**
For staff who don't live in Discord.

THE ASK

Three things.

- A** Greenlight rollout to all SCD machines next semester.
- B** Sponsor \$X for hosting + a paid student maintainer.
- C** Endorse Reserv as a template for other UIUC maker spaces.